

46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS **DT: 27 /03 /2026**
Time: 3 Hrs **GTM - 06** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. Match the physical quantities given in column I with dimensions expressed in column II in tabular form [M = mass, L= length , T= time , Q= charge]

Column I		Column II	
(P)	Angular Momentum	(i)	ML^2T^{-2}
(Q)	Latent Heat	(ii)	ML^2Q^{-2}
(R)	Torque	(iii)	ML^2T^{-1}
(S)	Capacitance	(iv)	$ML^3T^{-1}Q^{-2}$
(T)	Inductance	(v)	$M^{-1}L^{-2}T^2Q^2$
(U)	Resistivity	(vi)	L^2T^{-2}

Choose correct option from the below

For More Join: @IITJEE_Advanc

A) P—iii ,Q—vi ,R—i ,S-v,T—ii, U—iv

B) P—iii,Q—vi ,R—ii ,S-v T—i ,U—iv

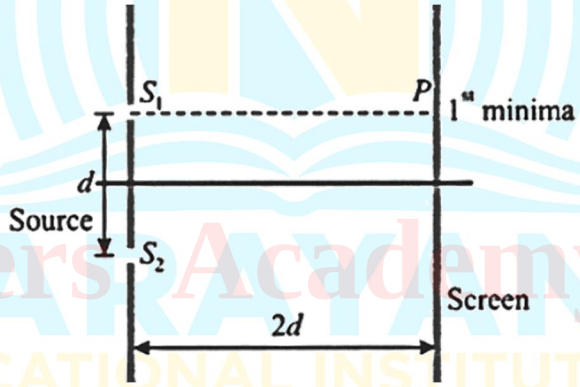
C) P—iii, Q—vi ,R—I ,S-v ,T—iv ,U—ii

D) P—iii,Q—v, R—I , S-vi ,T—ii, U—iv

2. A large number of water drops, each of radius r, combine to have a drop of radius R. If the surface tension is T and mechanical equivalent of heat is J, the rise in heat energy per unit volume will be

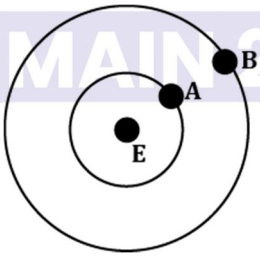
(A) $\frac{2T}{J}\left(\frac{1}{r}-\frac{1}{R}\right)$ (B) $\frac{2T}{rJ}$ (C) $\frac{3T}{rJ}$ (D) $\frac{3T}{J}\left(\frac{1}{r}-\frac{1}{R}\right)$

3. Consider a Young's double slit experiment as shown in figure. What should be the slit separation d in terms of wavelength λ such that the first minima occurs directly in front of the slit (S_1) ?



(A) $\frac{\lambda}{2\left(5-\sqrt{2}\right)}$ (B) $\frac{\lambda}{\left(5-\sqrt{2}\right)}$ (C) $\frac{\lambda}{\left(\sqrt{5}-2\right)}$ (D) $\frac{\lambda}{2\left(\sqrt{5}-2\right)}$

4. Two satellites A and B of masses 200 kg and 400 kg are revolving round the earth at height of 600 km and 1600 km respectively. If T_A and T_B are the time periods of A and B respectively then the value of $T_B - T_A$



[Given : radius of earth = 6400 km, mass of the earth = 6×10^{24} kg]

(A) 1.29×10^3 S (B) 3.33×10^2 S (C) 4.24×10^3 S (D) 4.24×10^2 S

For More Join: @IITJEE_Advanc

5. A plane electromagnetic wave travels in free space along the x-direction. The electric field component of the wave at a particular point of space and time is E = 6 V m^{-1} along y-direction. Its corresponding magnetic field component, B would be

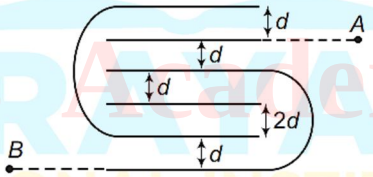
(A) 6×10^{-8} T along z-direction (B) 6×10^{-8} T along *x*-direction

(C) 2×10^{-8} T along z-direction (D) 2×10^{-8} T along y-direction

6. A block moving horizontally on a smooth surface with a speed of 40 m/s splits into two parts with masses in the ratio of 1:2. If the smaller parts move at 60 m/s in the same direction, then the fractional change in kinetic energy is

(A) $\frac{1}{3}$ (B) $\frac{2}{3}$ (C) $\frac{1}{4}$ (D) $\frac{1}{8}$

7. Find equivalent capacitance between points A and B. [Assume each conducting plate is having same dimensions and neglect the thickness of the plate, $\frac{\varepsilon_0 A}{d} = 7$ μF, where *A* is area of plates]

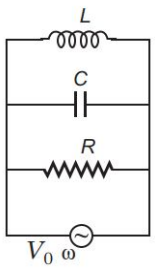


(A) 7 μF (B) 11 μF (C) 12 μF (D) 15 μF

8. According to Bohr model, magnetic field at centre (at the nucleus) of a hydrogen atom due to motion of electron in the nth orbit is proportional to

(A) $1/n^3$ (B) $1/n^5$ (C) n^5 (D) n^3

9. Consider an L-C-R circuit as shown in figure with an AC source of peak value V_0 and angular frequency ω. Then, the peak value of current through the AC source is



For More Join: @IITJEE_Advanc

(A) $\dfrac{V_0}{\sqrt{\dfrac{1}{R^2}+\left(\omega L-\dfrac{1}{\omega C}\right)^2}}$

(B) $V_0\sqrt{\left[\dfrac{1}{R^2}+\left(\omega C-\dfrac{1}{\omega L}\right)^2\right]}$

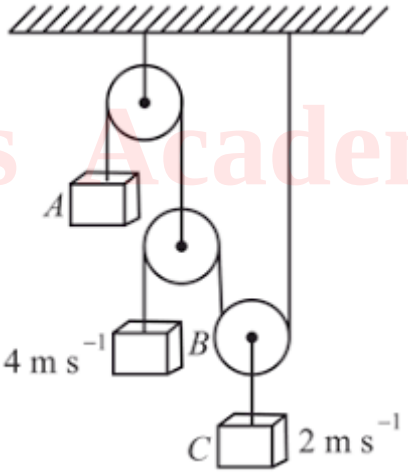
(C) $\dfrac{V_0}{\sqrt{R^2+\left(\omega L-\dfrac{1}{\omega C}\right)^2}}$

(D) None of these

10. Power applied to a particle varies with time as $P = [3t^2 - 2t + 1]$ watts. Where t is time in seconds. Then the change in kinetic energy of particle between time t = 2s to t = 4s is

(A) 46 J (B) 52 J (C) 92 J (D) 104 J

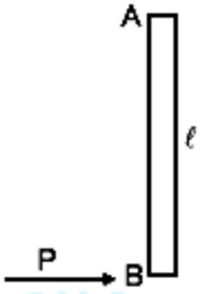
11. In the pulley system shown in figure, block C is going up at 2 m/s and block B is going up at 4 m/s, then the velocity of block A on the string shown in figure, is equal to :



(A) 2 m/s ↓ (B) 4 m/s ↓ (C) 6 m/s ↓ (D) 8 m/s ↓

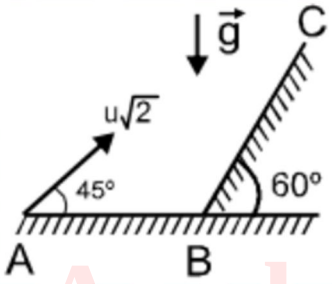
12. A uniform rod AB of mass m and length *l* at rest on a smooth horizontal surface. An impulse P is applied to the end B. The time taken by the rod to turn through a right angle is:

For More Join: @IITJEE_Advanc



(A) $\frac{2\pi ml}{P}$ (B) $\frac{\pi ml}{3P}$ (C) $\frac{\pi ml}{12P}$ (D) $\frac{2\pi ml}{3P}$

13. A particle is projected from point 'A' with velocity $u\sqrt{2}$ at an angle of 45° with the horizontal as shown in the figure. It strikes the inclined plane BC at right angle. The velocity of the particle just before the collision with the inclined is:



(A) $\frac{\sqrt{3}u}{2}$ (B) $\frac{u}{2}$ (C) $\frac{2u}{\sqrt{3}}$ (D) u

14. A jet of gas consists of molecules of mass m, speed *v* and number density n all moving co-linearly. This jet hits a wall at an angle θ to the normal. The pressure exerted on the wall by the jet assuming elastic collision will be:

(A) $p = 2mnv^2\cos^2\theta$ (B) $p = 2mnv^2\cos\theta$

(C) $p = \sqrt{(3/2)}\,mnv\,\cos^2\theta$ (D) $p = mnv^2$

15. A spherical planet of radius R has a uniform density ρ and does not rotate. If the planet is made up of some liquid, the pressure at point r from the center is:

(A) $\frac{4\pi\rho^2 G}{3}\left(R^2 - r^2\right)$ (B) $\frac{4\pi\rho G}{3}\left(R^2 - r^2\right)$

(C) $\frac{2\pi\rho^2 G}{3}\left(R^2 - r^2\right)$ (D) $\frac{\rho G}{2}\left(R^2 - r^2\right)$

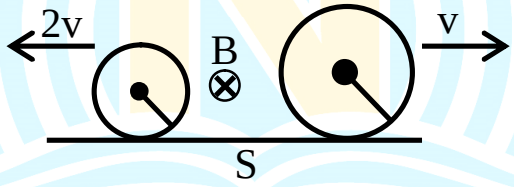
For More Join: @IITJEE_Advanc

16. A small loop of wire of area A = 0.01 m^2, N = 40 turns and resistance R = 20 Ω is initially kept in a uniform magnetic field B in such a way that the field is normal to the loop. When it is pulled out of the magnetic field, a total charge of Q = 2×10^{-5} C flows through the coil. The magnitude of magnetic field B is:

(A) 1×10^{-3} T (B) 4×10^{-3} T

(C) zero (D) unobtainable, as the data is insufficient

17. Two conducting rings of radii r and 2r move in opposite directions with velocities 2v and v respectively on a conducting surface S. There is a uniform magnetic field of magnitude B perpendicular to the plane of the rings . The potential difference between the highest points of the two rings is



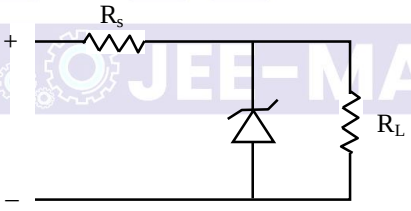
(A) zero (B) 2rvB (C) 4rv B (D) 8rv B

18. Two electric dipoles P_1 and P_2 are placed at (0, 0, 0) and (1, 0, 0) respectively with both of them pointing in the +z direction. Without changing the orientations of the dipoles P_2 is moved to (0, 2, 0). The ratio of the electrostatic potential energy of the dipoles after moving to that before moving is:

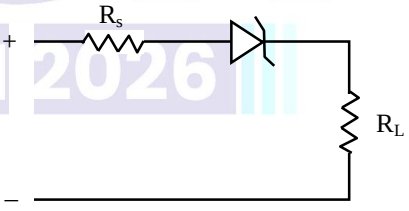
(A) $\frac{1}{16}$ (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) $\frac{1}{8}$

19. A zener diode is to be used as a voltage regulator. Identify the correct set up

A)

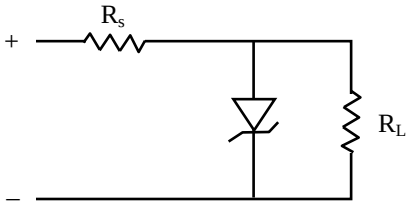


B)

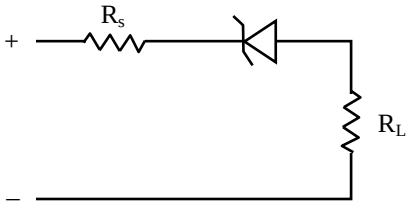


For More Join: @IITJEE_Advanc

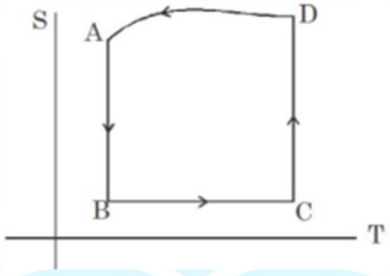
C)



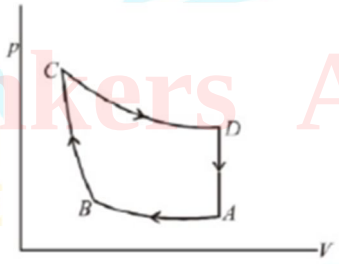
D)



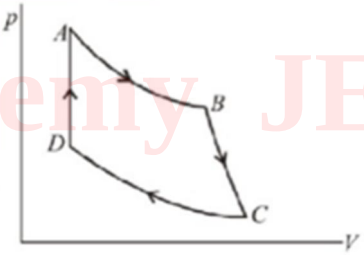
20. An ideal gas engine is run according to the cycle shown in the S-T diagram below, where the process from D to A is known to be isochoric (i.e., maintaining V = constant). The corresponding cycle in the P-V diagram will most closely resemble: ($S \rightarrow$ entropy, $T \rightarrow$ temperature, $P \rightarrow$ pressure, $V \rightarrow$ volume)



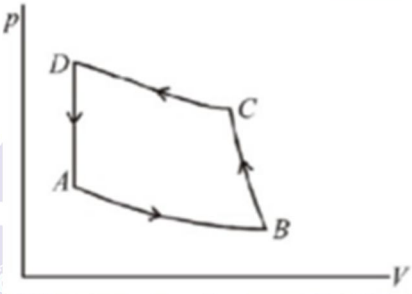
(A)



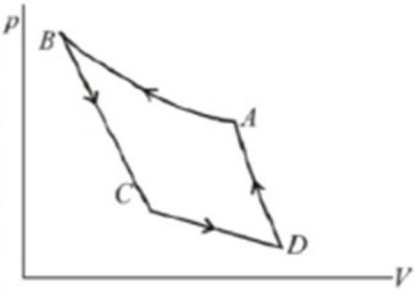
(B)



(C)



(D)



For More Join: @IITJEE_Advanc

SECTION – II
(Numerical Value Answer Type)

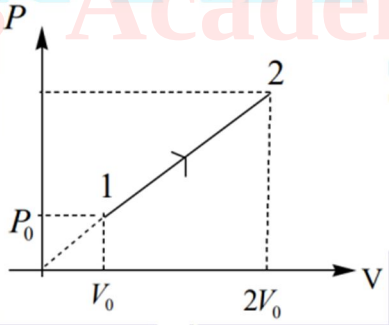
This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

21. An inclined plane is bent in such a way that the vertical cross-section is given by $y = \frac{x^2}{4}$ (x and y are in meter) where y is in vertical and x in horizontal direction.

If the upper surface of this curved plane is rough with coefficient of friction $\mu = 0.5$, the maximum height in cm at which a stationary block will not slip downward is _____ cm.

22. A convex lens made of glass (refractive index $=1.5$) has focal length 24 cm in air. When it is totally immersed in water (refractive index $=1.33$), its focal length changes to (in cm)

23. n moles of a monoatomic ideal gas undergoes a thermodynamic process as shown in figure from state 1 to state 2. Then the amount of heat supplied to the gas in this process is xP_0V_0 then value of 'x' is



24. The bulk modulus of rubber is 9.1×10^8 N/ m^2. To what depth(in meter) a rubber ball be taken in a lake so that its volume is decreased by 0.1%? $\left(g = 10 m/s^2\right)$

25. The time period of oscillation of a simple pendulum is given by $T = 2\pi\sqrt{\frac{l}{g}}$.

The length of the pendulum is measured as $l = 10 \pm 0.1$ cm and the time period as T = 0.5 ±0.02 s. Determine percentage error in the value of g.

For More Join: @IITJEE_Advanc

CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

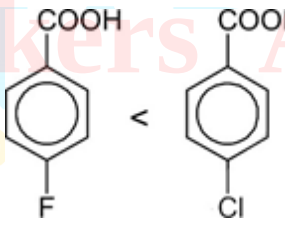
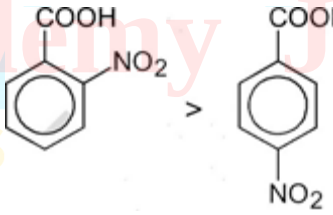
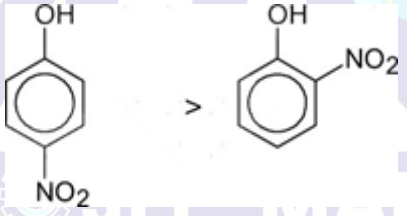
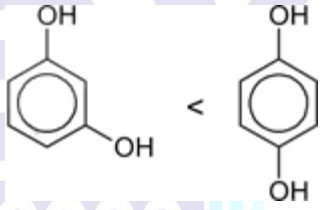
26. In the following molecular/ions $O_2^+, O_2^-, C_2^{-2}, N_2^{-4}$. Which one is paramagnetic and has highest bond length?

A) O_2^+ B) C_2^{-2} C) O_2^- D) N_2^{-4}

27. Difference between the radii of 2nd and 3rd orbits of He^+ is ΔR_1. The difference between the radii of 3rd and 4th orbits of Li^{+2} is ΔR_2. Find $\Delta R_1 / \Delta R_2$

A) $\frac{21}{10}$ B) $\frac{10}{21}$ C) $\frac{14}{15}$ D) $\frac{15}{14}$

28. Acidic strength order of compounds is given below. Identify incorrect option.

A)		B)	
C)		D)	

For More Join: @IITJEE_Advanc

29. Which of the following statement is incorrect?

A) Amylopectin is insoluble in water

B) Fructose is reducing sugar

C) Cellulose is the polymer of D-glucose

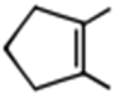
D) D-Ribose sugar is present in DNA

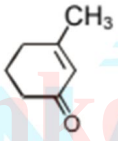
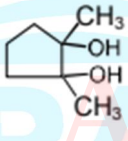
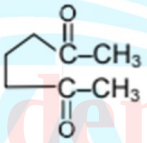
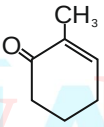
30. Which of the following complex compound show ionization, linkage and geometrical isomerism?

(I) $[Co(NH_3)_5(NO_2)]Br$; (II) $[Co(NH_3)_5(H_2O)](NO_2)_2$

(III) $[Pt(NH_3)_4][Pt(SCN)_4]$ (IV) $[Cr(NH_3)_4(NO_2)_2](NO_3)$

A) III only B) II and IV only C) IV only D) III and IV only

31.  $\xrightarrow[KMnO_4]{\text{cold.dil.aq.}}(X)\xrightarrow{HIO_4[O]}(Y)\xrightarrow{OH^-/\Delta}(Z)$, The product Z of the above reaction is

A)  B)  C)  D) 

32. Molecular formula of P and Q is C_2H_4O and C_3H_6O. If P and Q both give (+ve) $NaHSO_3$ test and only P gives haloform test. Choose the INCORRECT option:

$P+Q\xrightarrow[2.\Delta]{1.NaOH}$ Product

(A) Total 4 products will be obtained excluding stereoisomers.

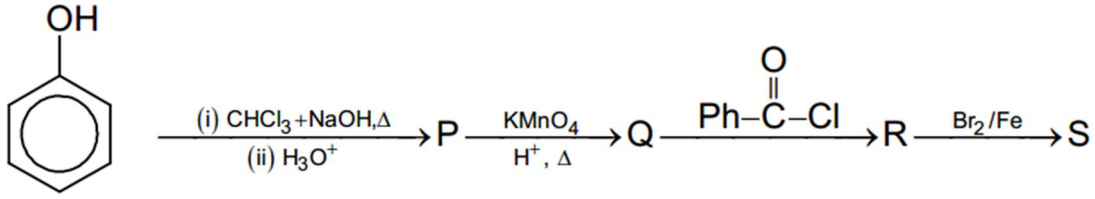
(B) Total 4 products will be obtained including stereoisomers.

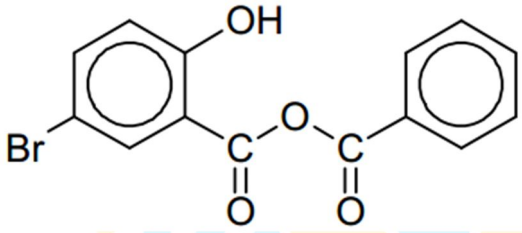
(C) Reaction is known as cross-Aldol condensation.

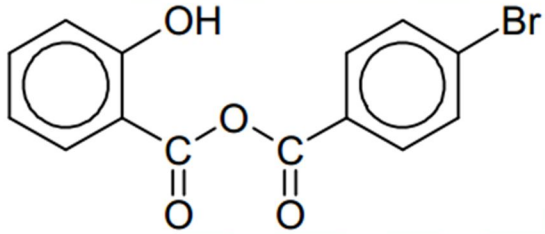
(D) All products obtained will give (+ve) Tollen't test as well as Fehling test.

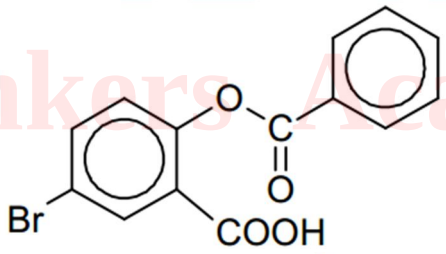
For More Join: @IITJEE_Advanc

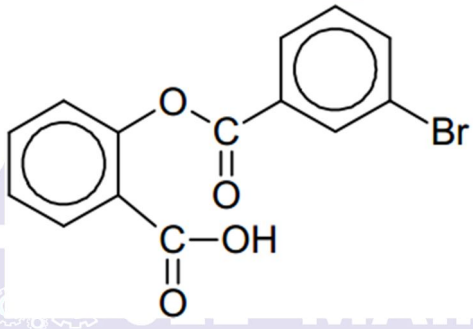
33. The major product 'S' obtained in the following reaction is



(A)


(B)


(C)


(D)


34. Compare van't Hoff factor for compounds CH_3CH_2OH (A) and Ph_3COH (B) in cold conc. H_2SO_4 determined from measurement of freezing point.

(A) $i_A > i_B$ (B) $i_A < i_B$ (C) $i_A = i_B$ (D) $i_A < i_B < 1$

For More Join: @IITJEE_Advanc

35. Glucose is added to 1 litre water to such an extent that $\Delta T_f / K_f$ becomes equal to 1/1000, the wt. of glucose added is

(A) 180 g (B) 18 g (C) 1.8 g (D) 0.18 g

36. The rate constant of a second order reaction A → B is k_2. If the initial concentration of the reactant is x_o and the concentration of the product at time t is a, then a linear function of t with the slope $k_2 x_o$ is:

(A) $\ln\left(\frac{a}{x_o - a}\right)$ (B) $\left(\frac{a}{x_o(x_o - a)}\right)$

(C) $\left(\frac{a}{x_o - a}\right)$ (D) $\ln\left(\frac{a}{x_o(x_o - a)}\right)$

37. The increasing order of bond angles in OF_2, OCl_2, ClO_2 and H_2O is ____.

(A) $OF_2 < H_2O < OCl_2 < ClO_2$ (B) $H_2O < F_2O < Cl_2O < ClO_2$

(C) $OF_2 < Cl_2O < H_2O < ClO_2$ (D) $ClO_2 < Cl_2O < OF_2 < H_2O$

38. The enthalpy change on freezing of one mole of water at 10^0C to -10^0C is

(Given data: $\Delta_{fus}H = 6.03$ KJ/mol at 0^0C, C_P of $H_2O_{(l)} = 75.3$ J/mol K, $C_P H_2O_{(s)} = 36.8$ J/mol/ K)

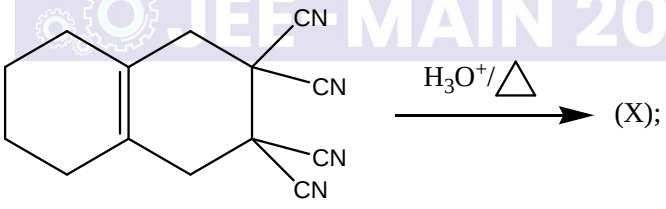
A) $-7151J$ B) $-4917J$ C) $-1127.03J$ D) $-6415J$

39. Y on reaction with $Cl_2 / AlCl_3$ gives:

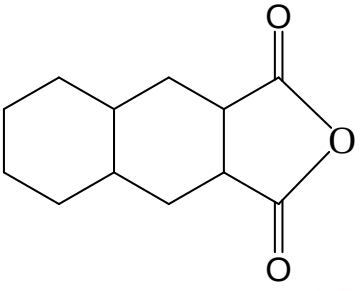
$$CaC_2 \xrightarrow{H_2O} 'X'_{(gas)} \xrightarrow[\text{Fe Tube}]{\text{Red Hot}} 'Y'_{(gas)}$$

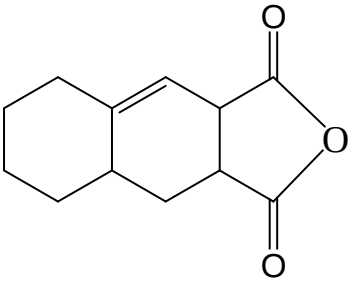
(A) C_2H_2 (B) $CH_2 = CH - Cl$ (C) C_6H_6 (D) $Ph - Cl$

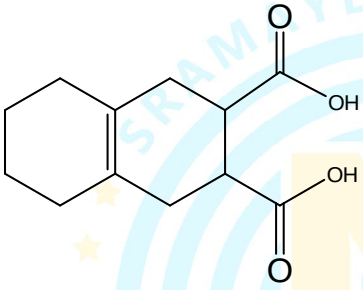
40. The product (X) is:

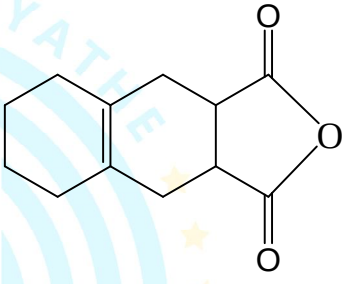


For More Join: @IITJEE_Advanc

(A) 

(B) 

(C) 

(D) 

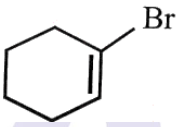
41. Chloro compound of Vanadium has only spin magnetic moment of 1.73 BM. This Vanadium chloride has the formula:

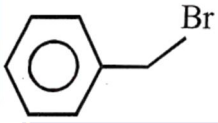
A) VCl_2 B) VCl_4 C) VCl_3 D) VCl_5

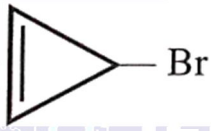
42. Number of Cl = O bonds in chlorous acid, chloric acid and perchloric acid respectively are:

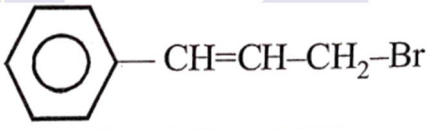
(A) 3, 1 and 1 (B) 4, 1 and 0 (C) 1, 1 and 3 (D) 1, 2 and 3

43. Compound from the following that will not produce precipitate on reaction with $AgNO_3$ is:

(A) 

(B) 

(C) 

(D) 

44. A sample of $CaCO_3$ and $MgCO_3$ weighed 2.21g is ignited to constant weight of 1.152 g. The composition of mixture is (Given, molar mass in g mol^{-1} $CaCO_3$: 100, $MgCO_3$: 84)

(A) 1.187 g $CaCO_3$ + 1.187 g $MgCO_3$ (B) 1.023 g $CaCO_3$ + 1.187 g $MgCO_3$

(C) 1.187 g $CaCO_3$ + 1.023 g $MgCO_3$ (D) 1.023 g $CaCO_3$ + 1.023 g $MgCO_3$

For More Join: @IITJEE_Advanc

45. Given below are two statements.

Statement I: Since fluorine is more electronegative than nitrogen, the net dipole moment of NF_3 is

greater than NH_3.

Statement II: In NH_3 the orbital dipole due to lone pair and the dipole moment of NH bonds are in

opposite direction, but in NF_3 the orbital dipole due to lone pair and dipole moments of N—F bonds

are in same direction.

(A) Statement I is false but Statement II is true.

(B) Statement I is true but Statement II is false.

(C) Both Statement I and Statement II are true.

(D) Both Statement I and Statement II are false.

SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

46. Enthalpy change (in kcal) for a hypothetical reaction

$AB_2(g) \rightarrow A^+(g) + B^-(g) + B(g)$ is x kcal/mole. Find out x/100?

(Average bond enthalpy of A – B = 100 kcal/mole, 1st ionization enthalpy of A(g) = 250 Kcal/mole. Electron gain enthalpy of B(g) = - 150 kcal/mole)

47. In a system, where $\Delta U = -15kJ$, a piston expanded against a external pressure of 1.2 atm giving a change in volume of 75 lit. What was the magnitude of change in heat energy of this system (in kJ)?

(take 1lt atm = 100 J)

48. A volume of 500 ml of 0.1 M $CuSO_4$ solution is electrolysed for 5 min at a current of 0.161A. If Cu is produced at one electrode and oxygen at the other, the approximate pH of the final solution is

For More Join: @IITJEE_Advanc

49. The electrode potential of the following cell is $-x$ at 300 K.

$$Pt(s)\,|\,H_2(g)(4\ atm)\,/\,CH_3COOH(aq)(10^{-1}\ M)\,||\,NaOH(aq)(10^{-3}\ M)\,/\,H_2(g)(1\ atm)\,/\,Pt(s).$$

The value of 1000 x is_____.

[Given: $\frac{2.303RT}{F}$ at 300 K = 0.06, log2 = 0.3, K_a for acetic acid is 10^{-5}].

50. A compound (X) with molar mass 108 g mol^{-1} undergoes acetylation to give product with molar mass 192 g mol^{-1}. The number of amino groups in the compound (X) is _____

MATHEMATICS MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. Let $f(x)$ be a polynomial of least degree such that $\lim\limits_{x\to 0}\left(1+\frac{x^2+f(x)}{x^2}\right)^{\frac{1}{x}}=e^2$ and

$$h(x)=\begin{cases} f(x)+b & x\le 0 \\ \frac{e^x-e^{x\cos x}}{x+\sin x} & x>0 \end{cases}$$ is continuous at $x=0$ then b is equal to:

(A) 0 (B) 1 (C) 4 (D) 5

52. An urn contains 4 green balls and 6 blue balls. A second urn contains 16 green balls and n blue balls. A single ball is drawn at random from each urn and probability that both balls are of same colour is $\frac{29}{50}$. If first two drawn balls (one from each urn) are both green and probability of further two drawn balls (one from each urn and without replacing first two balls) are same colour is p, then p is equal to:

(A) $\frac{5}{159}$ (B) $\frac{96}{159}$ (C) $\frac{101}{159}$ (D) $\frac{91}{159}$

For More Join: @IITJEE_Advanc

53. In the solution of $\left(1+2e^{\frac{x}{y}}\right)dx+2e^{\frac{x}{y}}\left(1-\frac{x}{y}\right)dy=0$ is $\left(x+\lambda ye^{\frac{x}{y}}\right)=c;$(c is an arbitrary constant), they λ is:

(A) 1 (B) 2 (C) 3 (D) 6

54. The points (5,12) and (24,7) are the foci of an ellipse passing through origin. The eccentricity of an hyperbola is $\sqrt{2}$ times the greatest integer less than reciprocal of eccentricity of given ellipse if the distance between foci of the hyperbola is 16, then equation of hyperbola is:

(A) $x^2-y^2=32$ (B) $x^2-y^2=16$ (C) $\frac{x^2}{16}-\frac{y^2}{9}=1$ (D) $\frac{x^2}{32}-\frac{y^2}{16}=1$

55. Given that variance of 20 observations is 5,when each of those observation is multiplied by -2, then new variance is,

(A) 10 (B) 30 (C) -20 (D) 20

56. The line L given by $\frac{x-2}{2}=\frac{y-1}{b}=\frac{z+1}{c}$ passes through the point (1, 2, 3). Another line K is parallel to line L and has the equation $\frac{x+2}{a}=\frac{y-3}{2}=\frac{z+4}{d}$. Then the distance between line L and K is

A) $\frac{\sqrt{297}}{3}$ B) $\frac{\sqrt{243}}{3}$ C) $\frac{\sqrt{272}}{9}$ D) $\frac{\sqrt{291}}{9}$

57. Let $f(x)$ be a function continuous on [1,2] and differentiable on (1,2) satisfying $f(1)=2,\ f(2)=3$ and $f'(x)\geq 1\ \forall x\in(1,2)$ Define $g(x)=\int_1^x f(t)dt\ \forall x\in[1,2]$ then the greatest value of $g(x)$ on [1,2] is

A) 3 B) 5 C) $\frac{5}{2}$ D) $\frac{3}{2}$

58. Let any double ordinate PNP' of the hyperbola $\frac{x^2}{25}-\frac{y^2}{16}=1$ be produced on both sides to meet the asymptotes in Q and Q'. Then $PQ.P'Q$ is equal to

A) 25 B) 16 C) 41 D) none of these

For More Join: @IITJEE_Advanc

59. Locus of the point from where 3 normals are drawn to the parabola $y^2 = 4x$ such that two of them are perpendicular is

A) $y^2 = (x-1)$ B) $y^2 = (x-3)$ C) $y^2 = (x+3)$ D) $y^2 = (x+1)$

60. All words, with or without meaning, are made using all the letters of the word MONDAY. These words are written as in a dictionary with serial numbers. The serial number of the word MONDAY is

(A) 327 (B) 328 (C) 324 (D) 326

61. A tangent to the hyperbola $\frac{x^2}{4} - \frac{y^2}{2} = 1$ meet x-axis at P and y –axis at Q, lines PR are QR are drawn such that OPRQ is a rectangle where O is origin then R lies on

A) $\frac{4}{x^2} + \frac{2}{y^2} = 1$ B) $\frac{2}{x^2} - \frac{4}{y^2} = 1$ C) $\frac{2}{x^2} + \frac{4}{y^2} = 1$ D) $\frac{4}{x^2} - \frac{2}{y^2} = 1$

62. The value of $\sum_{n=1}^{\infty} \sin^{-1}\left[\frac{2n+1}{n(n+1)\left[\sqrt{n^2+2n}+\sqrt{n^2-1}\right]}\right]$ is equal to

A) $\frac{\pi}{4}$ B) $\frac{\pi}{6}$ C) $\frac{3\pi}{4}$ D) $\frac{\pi}{2}$

63. If α, β and γ are roots of $x^3 - 2x^2 + 6x - 1 = 0$ the value of the expression

$$\alpha\left(\frac{\alpha^2+\alpha+1}{\alpha^2-\alpha+1}\right) + \beta\left(\frac{\beta^2+\beta+1}{\beta^2-\beta+1}\right) + \gamma\left(\frac{\gamma^2+\gamma+1}{\gamma^2-\gamma+1}\right)$$

A) 1 B) 6 C) 7 D) 8

64. The value of definite Integral $\int_{\frac{-1}{\sqrt{3}}}^{\frac{1}{\sqrt{3}}} \frac{\cos^{-1}\left(\frac{2x}{1+x^2}\right) + \tan^{-1}\left(\frac{2x}{1-x^2}\right)}{e^x+1} dx$ is equal to

A) $\frac{\pi}{2\sqrt{3}}$ B) $\frac{\pi}{\sqrt{3}}$ C) $\frac{\pi}{4\sqrt{3}}$ D) $\frac{\pi}{3\sqrt{3}}$

65. The function $f(x) = |x^2 - 2x - 3| . e^{|9x^2-12x+4|}$ is not differentiable at exactly:

A) four points B) three points

C) two points D) one points

For More Join: @IITJEE_Advanc

66. Let $f:R\to R, g:R\to R$ and $h:R\to R$ be differentiable functions such that $f(x)=x^3+3x+2, g(f(x))=x$ and $h(g(x))=f(x)$ for all $x\in R$. Then

A) $g'(2)=\frac{1}{15}$ B) $h'(1)=666$

C) $h(0)=166$ D) $h(g(3))=36$

67. If $I_1=\int_0^1 \frac{x^{7/2}(1-x)^{9/2}}{30}dx$ and $I_2=\int_0^1 \frac{x^{7/2}(1-x)^{9/2}}{(x+5)^{10}}dx$ and $\frac{I_1}{I_2}=5a^3\sqrt{a}$ where $a\in N$, Then the value of a is

A) 24 B) 26 C) 28 D) 30

68. If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are such that $[\vec{a}\vec{b}\vec{c}]=1$, $\vec{c}=\lambda(\vec{a}\times\vec{b})$, $|\vec{a}|=\sqrt{2}$, $|\vec{b}|=\sqrt{3}$ and $|\vec{c}|=\frac{1}{\sqrt{3}}$, then the angle between $\vec{a}$ and $\vec{b}$ is

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{6}$

69. $\int \frac{1}{\left[(x-1)^3(x+2)^5\right]^{1/4}}dx$ is equal to

(A) $\frac{4}{3}\left(\frac{x-1}{x+2}\right)^{1/4}+C$ (B) $\frac{4}{3}\left(\frac{x+2}{x-1}\right)^{1/4}+C$

(C) $\frac{1}{3}\left(\frac{x-1}{x+2}\right)^{1/4}+C$ (D) $\frac{1}{3}\left(\frac{x+2}{x-1}\right)^{1/4}+C$

70. Let $f(x)=$ minimum $\left(x+1,\sqrt{1-x}\right)$ for all $x\le 1$. Then the area bounded by $y=f(x)$ and the x-axis is

(A) $\frac{7}{3}$ sq.units (B) $\frac{1}{6}$ sq.units (C) $\frac{11}{6}$ sq.units (D) $\frac{7}{6}$ sq.units

For More Join: @IITJEE_Advanc

SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

71. In the expansion $\frac{\left(x^5+1\right)\left(1+\frac{5}{x}+\frac{10}{x^2}+\frac{10}{x^3}+\frac{5}{x^4}+\frac{1}{x^5}\right)}{\left(x^4-x^3+x^2-x+1\right)}$ $x\neq 0$, the sum of the coefficient of x^{-3} and constant terms is _____.

72. Consider the chords of the parabola $y^2 = 4x$ which touches the hyperbola $x^2 - y^2 = 1$, the locus of the point of intersection of the tangents drawn to the parabola at the extremities of such chords is a conic section having latus rectum λ, the value of λ is

73. α and β are complex numbers satisfying $|\alpha-1+i|=2$ and $|\beta-4+5i|=3$ and $2|\beta|_{max}-3|\alpha|_{max}=\sqrt{k_1}-\sqrt{k_2}$ then $\left[\frac{k_1}{k_2}\right]$ is equal to [.] greatest integer function

74. Let $\vec{a}=2\hat{i}+\hat{j}-2\hat{k}$ and $\vec{b}=\hat{i}+\hat{j}$. If c is a vector such that $\vec{a}.\vec{c}=|\vec{c}|,|\vec{c}-\vec{a}|=2\sqrt{2}$ and the angle between $\vec{a}\times\vec{b}$ and $\vec{c}$ is 30°, then $\frac{2}{3}|(\vec{a}\times\vec{b})\times\vec{c}|$ is equal to

75. If the Locus of the image of the Point $\left(\lambda^2,2\lambda\right)$ in the mirror $x-y+1=0$ (where λ is a parameter) is $(x+a)^2=b(y-c)$ where $a,b,c\in I$ then the value of $\left(\frac{a+b}{c+b}\right)$ is equal to :

For More Join: @IITJEE_Advanc